

Types of Surds

- Pure surds** A surd which has only 1 as a rational factor is called a pure surd.
e.g. $\sqrt{2}$, $\sqrt[3]{2}$, $\sqrt[4]{3}$ all are pure surds.
- Mixed surds** A surd which is not a pure surd or has factor other than unity is called a mixed surd.
e.g. $2\sqrt{3}$, $5\sqrt[3]{12}$, $\frac{4}{3}\sqrt[3]{35}$ are mixed surds.
- Like and Unlike surds** When the radicands of two surds are same, then those are known as like surds.
e.g. $6\sqrt{3}$ and $2^2 \cdot \sqrt{3}$.
When radicands of two surds are different, then they are called unlike surds. e.g. $4\sqrt{7}$ and $6\sqrt{5}$.

Comparison of Two Surds

- If two surds are of the same order, then the one whose radicand is larger, is the larger of the two.
e.g. $\sqrt{17} > \sqrt{13}$, $\sqrt[3]{21} > \sqrt[3]{16}$
- If any two surds have different orders are to be compared, then we will first reduce them to the same but smallest order and then compare them.

Note Let $\sqrt[n]{x}$ be a surd of order n .
Then, $\sqrt[n]{x} = \sqrt[m]{x^{m/n}}$ i.e. $x^{1/n} = (x^{m/n})^{1/m}$ is a surd of order m .

EXAMPLE 9. Convert $\sqrt{2}$ into a surd of order 4.

- a. $\sqrt[4]{2}$ b. $\sqrt[4]{8}$ c. $\sqrt[4]{4}$ d. None of these

Sol. c. $\sqrt{2} = 2^{1/2} = (2^{4/2})^{1/4} = (2^2)^{1/4} = \sqrt[4]{4}$

EXAMPLE 10. Consider the following in respect of the numbers $\sqrt{2}$, $\sqrt[3]{3}$, and $\sqrt[6]{6}$. ☞ 2014 (I)

- $\sqrt[6]{6}$ is the greatest number.
- $\sqrt{2}$ is the smallest number.

Which of these statements is/are correct?

- a. Only I b. Only II c. Both I and II d. Neither I nor II

Sol. d. Taking LCM of 2, 3 and 6 = 12

$$\text{Now, } \sqrt{2} = 2^{1/2} = 2^{6/12} = \sqrt[12]{2^6} = \sqrt[12]{64}$$

$$\sqrt[6]{6} = (6)^{1/6} = (6)^{2/12} = \sqrt[12]{6^2} = \sqrt[12]{36}$$

$$\sqrt[3]{3} = 3^{1/3} = 3^{4/12} = \sqrt[12]{81}$$

So, neither I nor II is correct.

Operations on Surds

Addition and Subtraction of Surds

Addition or subtraction takes place only between like surds, using laws of numbers. e.g.

- $2\sqrt{3} + 4\sqrt{3} + \sqrt{5} = 6\sqrt{3} + \sqrt{5}$
- $6\sqrt{5} - 3\sqrt{2} + 7\sqrt{5} + 4\sqrt{2} = (6\sqrt{5} + 7\sqrt{5}) + (4\sqrt{2} - 3\sqrt{2})$
 $= 13\sqrt{5} + \sqrt{2}$

Multiplication and Division of Surds

Suppose given surds are $\sqrt[p]{a}$, $\sqrt[q]{b}$ and $\sqrt[r]{c}$, then first find the LCM of p , q and r to make the radical powers of the surds same and then their multiplication and division can be done using following rules.

$$(i) \sqrt[p]{x} \times \sqrt[q]{y} = \sqrt[pq]{xy} \quad (ii) \frac{\sqrt[p]{x}}{\sqrt[q]{y}} = \sqrt[pq]{\frac{x}{y}}$$

EXAMPLE 11. Simplify $\sqrt[6]{12} \div \sqrt{3} \cdot \sqrt[3]{2}$.

- a. $\sqrt[3]{3}$ b. $\sqrt{3}$
c. $\sqrt[3]{2}$ d. $\sqrt[3]{\frac{1}{3}}$

Sol. d. Here, the order of $\sqrt{3}$, $\sqrt[3]{2}$ is 2 and 3, respectively.

So, LCM 2, 3 and 6 = 6

$$\sqrt{3} = 3^{1/2} = (3^{6/2})^{1/6} = (3^3)^{1/6} = (27)^{1/6} = \sqrt[6]{27}$$

$$\sqrt[3]{2} = (2)^{1/3} = (2^{6/3})^{1/6} = (2^2)^{1/6} = \sqrt[6]{4}$$

$$\begin{aligned} \text{So, } \frac{\sqrt[6]{12}}{\sqrt{3} \cdot \sqrt[3]{2}} &= \frac{\sqrt[6]{12}}{\sqrt[6]{27} \cdot \sqrt[6]{4}} = \frac{\sqrt[6]{12}}{\sqrt[6]{27 \times 4}} = \sqrt[6]{\frac{12}{27 \times 4}} = \sqrt[6]{\frac{1}{9}} \\ &= \left[\left(\frac{1}{3} \right)^2 \right]^{1/6} = \left[\frac{1}{3} \right]^{2/6} = \left[\frac{1}{3} \right]^{1/3} = \sqrt[3]{\frac{1}{3}} \end{aligned}$$

Rationalisation of Surds

Process of converting surd into rational number is called as rationalisation of surds.

When the product of two surds is a rational number, then each surd is called a rationalising factor of each other.

$$\text{e.g. } \sqrt[3]{25} \times \sqrt[3]{5} = \sqrt[3]{25 \times 5} = \sqrt[3]{125} = \sqrt[3]{5^3} = 5$$

So, $\sqrt[3]{25}$ and $\sqrt[3]{5}$ are rationalising factor of each other.

Rationalising factors of a given surd

- Rationalising factor of $\frac{1}{\sqrt{a}} = \sqrt{a}$
- Rationalising factor of $\frac{1}{\sqrt{a} \pm \sqrt{b}} = \sqrt{a} \mp \sqrt{b}$

EXAMPLE 12. $\frac{\sqrt{5} + \sqrt{3}}{\sqrt{5} - \sqrt{3}} + \frac{\sqrt{5} - \sqrt{3}}{\sqrt{5} + \sqrt{3}}$ is equal to

- a. 16 b. 8 c. 4 d. $\sqrt{15}$

Sol. b.
$$\begin{aligned} &\frac{\sqrt{5} + \sqrt{3}}{\sqrt{5} - \sqrt{3}} + \frac{\sqrt{5} - \sqrt{3}}{\sqrt{5} + \sqrt{3}} \\ &= \frac{(\sqrt{5} + \sqrt{3})(\sqrt{5} + \sqrt{3})}{(\sqrt{5} - \sqrt{3})(\sqrt{5} + \sqrt{3})} + \frac{(\sqrt{5} - \sqrt{3})(\sqrt{5} - \sqrt{3})}{(\sqrt{5} + \sqrt{3})(\sqrt{5} - \sqrt{3})} \\ &= \frac{(\sqrt{5} + \sqrt{3})^2}{\sqrt{5}^2 - \sqrt{3}^2} + \frac{(\sqrt{5} - \sqrt{3})^2}{\sqrt{5}^2 - \sqrt{3}^2} = \frac{(\sqrt{5} + \sqrt{3})^2 + (\sqrt{5} - \sqrt{3})^2}{(\sqrt{5})^2 - (\sqrt{3})^2} \\ &= \frac{2[(\sqrt{5})^2 + (\sqrt{3})^2]}{5 - 3} = \frac{2(5 + 3)}{2} = 8 \end{aligned}$$

PRACTICE EXERCISE

- If $\sqrt{\frac{x}{49}} = \frac{4}{7}$, then the value of x is
(a) 9 (b) 25 (c) 16 (d) 8
- Which of the following cannot be a digit in the unit place of a perfect square?
(a) 1 (b) 5 (c) 7 (d) 0
- The number $\sqrt{0.0001}$ is
(a) a rational number less than 0.01
(b) a rational number
(c) an irrational number
(d) Neither a rational number nor an irrational number
- If m and n are natural numbers, then $\sqrt[n]{n}$ is
(a) always irrational
(b) irrational unless n is the m th power of an integer
(c) irrational unless m is the n th power of an integer
(d) irrational unless m and n are coprime
- What is that fraction which when multiplied by itself gives 227.798649?
(a) 15.093 (b) 15.099 (c) 14.093 (d) 9.0019
- If $x = \sqrt{3018 + \sqrt{36 + \sqrt{169}}}$, then the value of x is
(a) 55 (b) 44 (c) 63 (d) 42
- If $\sqrt[3]{\sqrt{x}} = x/5$, then x is equal to
(a) $\sqrt[5]{5}$ (b) $\sqrt[6]{5^5}$ (c) $\sqrt[5]{5^6}$ (d) $\sqrt{5}$
- What is the value of
 $\sqrt{29.16} + \sqrt{0.2916} + \sqrt{0.002916} + \sqrt{0.00002916}$?
(a) 5.9949 (b) 5.9894 (c) 5.9984 (d) 5.9994
- $(\sqrt{5})^{-5/2} \times (\sqrt{5})^{-3/2}$
(a) $\frac{1}{625}$ (b) $\frac{1}{25}$ (c) $\frac{1}{125}$ (d) None of these
- If $a = 3$, $b = 9$ and $c = 10$, then the value of
 $\sqrt{13+a} + \sqrt{112+b} + \sqrt{c-1}$ is
(a) 15 (b) 18 (c) 16 (d) 10
- If $\sqrt{0.9 \times 0.09 \times x} = 0.9 \times 0.09 \times \sqrt{z}$, then the value of $\frac{x}{z}$ is
(a) 0.081 (b) 0.810 (c) 0.81 (d) 8.09
- $\sqrt{2\sqrt{2\sqrt{2\sqrt{2\sqrt{2}}}}}$ is equal to
(a) 0 (b) 2 (c) 1 (d) $2^{31/32}$
- If $\sqrt{35} = 5.9160$, then the value of $\frac{\sqrt{7} + \sqrt{5}}{\sqrt{7} - \sqrt{5}}$ is
(a) 9.1060 (b) 10.9160 (c) 11.9160 (d) 12
- Find the value of
 $\sqrt{343 + \sqrt{307 + \sqrt{273 + \sqrt{241 + \sqrt{225}}}}}$
(a) 18 (b) 19 (c) $\sqrt{19}$ (d) $18 + \sqrt{3}$
- What is $\frac{1}{\sqrt{9} - \sqrt{8}} - \frac{1}{\sqrt{8} - \sqrt{7}} + \frac{1}{\sqrt{7} - \sqrt{6}} - \frac{1}{\sqrt{6} - \sqrt{5}}$
 $+ \frac{1}{\sqrt{5} - \sqrt{4}}$ equal to?
(a) 0 (b) 1 (c) 5 (d) $\frac{1}{3}$
- What is one of the square roots of $9 - 2\sqrt{14}$?
(a) $\sqrt{7} - \sqrt{3}$ (b) $\sqrt{6} - \sqrt{3}$ (c) $\sqrt{7} - \sqrt{5}$ (d) $\sqrt{7} - \sqrt{2}$
- If $p^x = r^y = m$ and $r^w = p^z = n$, then which one of the following is correct?
(a) $xw = yz$ (b) $xz = yw$
(c) $x + y = w + z$ (d) $x - y = w - z$
- If $a^x = b^y = c^z$ and $abc = 1$, then what is $xy + yz + zx$ equal to
(a) xyz (b) $x + y + z$ (c) 0 (d) 1
- If $a = \frac{\sqrt{5} + 1}{\sqrt{5} - 1}$ and $b = \frac{\sqrt{5} - 1}{\sqrt{5} + 1}$, then the value of
 $\left(\frac{a^2 + ab + b^2}{a^2 - ab + b^2}\right)$ is
(a) $3/4$ (b) $4/3$ (c) $3/5$ (d) $5/3$
- If $3\sqrt{5} + \sqrt{125} = 17.88$, then what will be the value of $\sqrt{80} + 6\sqrt{5}$
(a) 13.41 (b) 20.46 (c) 21.66 (d) 22.35
- If $N = 2^{0.15}$ and $N^b = 16$, then b is equal to
(a) $80/3$ (b) $5/3$ (c) 4 (d) $3/5$
- If $6^A = 2$, $6^B = 5$ and $6^Q = 15$, then Q equals to
(a) $B + 3$ (b) $5A + B$ (c) $B^2 - 2B$ (d) $B - A + 1$
- Arrange $\sqrt[4]{3}$, $\sqrt[6]{10}$, $\sqrt[12]{25}$ in descending order.
(a) $\sqrt[6]{10} > \sqrt[4]{3} > \sqrt[12]{25}$ (b) $\sqrt[12]{25} > \sqrt[4]{3} > \sqrt[6]{10}$
(c) $\sqrt[6]{10} > \sqrt[12]{25} > \sqrt[4]{3}$ (d) $\sqrt[4]{3} > \sqrt[12]{25} > \sqrt[6]{10}$
- If $(3.7)^x = (0.037)^y = 10000$, then what is the value of $\frac{1}{x} - \frac{1}{y}$?
(a) 1 (b) 2 (c) $\frac{1}{2}$ (d) $\frac{1}{4}$
- The greatest six digit number which is a perfect square is
(a) 998004 (b) 998006 (c) 998049 (d) 998001

- 26.** What is the smallest number by which 26244 must be divided to get a perfect cube?
(a) 4 (b) 6 (c) 36 (d) 16
- 27.** What is the smallest number that must be added to 1780 to make it a perfect square?
(a) 39 (b) 49 (c) 59 (d) 69
- 28.** A gardener plants 17956 trees in such a way that there are as many rows as there are trees in a row. The number of trees in a row are
(a) 136 (b) 164 (c) 134 (d) 166
- 29.** A group of student decided to collect as many paise from each member of the group as is the number of members. If the total collection amounts to ₹ 32.49 the number of members in the group, is
(a) 37 (b) 47 (c) 57 (d) 27
- 30.** A general arranges his soldiers in rows to form a perfect square. He find that in doing, so 60 soldiers are left out. If the total number of soldiers be 8160. Then, the number of soldiers in each row is
(a) 90 (b) 91 (c) 92 (d) 80
- 31.** A ball is dropped from a height 64 m above the ground and every time it hits the ground it rises to a height equal to half of the previous. What is the height attained after it hits the ground for the 16th time?
(a) 2^{-12} m (b) 2^{-11} m (c) 2^{-10} m (d) 2^{-9} m
- 32.** If $x = \frac{\sqrt{a+3b} + \sqrt{a-3b}}{\sqrt{a+3b} - \sqrt{a-3b}}$, then $3bx^2 - 2ax + 3b$ is equal to (given that, $b \neq 0$).
(a) 1 (b) 0 (c) ab (d) $2ab$
- 37.** If $16 \times 8^{n+2} = 2^m$, then m is equal to ☑ 2013 I
(a) $n + 8$ (b) $2n + 10$
(c) $3n + 2$ (d) $3n + 10$
- 38.** When a ball bounces, it rises to $\frac{2}{3}$ of the height from which it fell. If the ball is dropped from a height of 36 m, how high will it rise at the third bounce? ☑ 2013 I
(a) $10\frac{1}{3}$ m (b) $10\frac{2}{3}$ m (c) $10\frac{1}{3}$ m (d) $12\frac{2}{3}$ m
- 39.** The product of four consecutive natural numbers plus one is ☑ 2014 I
(a) a non-square
(b) always sum of two square numbers
(c) a square
(d) None of the above
- 40.** The difference of two consecutive cubes ☑ 2014 I
(a) is odd or even (b) is never divisible by 2
(c) is always even (d) None of these
- 41.** The square root of $\frac{(0.75)^3}{1 - 0.75} + [0.75 + (0.75)^2 + 1]$ is ☑ 2015 I
(a) 1 (b) 2 (c) 3 (d) 4
- 42.** What is $\frac{5 + \sqrt{10}}{5\sqrt{5} - 2\sqrt{20} - \sqrt{32} + \sqrt{50}}$ equal to? ☑ 2015 I
(a) 5 (b) $5\sqrt{2}$ (c) $5\sqrt{5}$ (d) $\sqrt{5}$
- 43.** If $x = \frac{91}{216}$, then the value of $3 - \frac{1}{(1-x)^{1/3}}$ is ☑ 2015 II
(a) $\frac{9}{5}$ (b) $\frac{5}{9}$ (c) $\frac{4}{9}$ (d) $\frac{4}{5}$
- 44.** Which one of the following is correct? ☑ 2015 II
(a) $\sqrt{2} < \sqrt[4]{6} < \sqrt[3]{4}$ (b) $\sqrt{2} > \sqrt[4]{6} > \sqrt[3]{4}$
(c) $\sqrt[4]{6} < \sqrt{2} < \sqrt[3]{4}$ (d) $\sqrt[4]{6} > \sqrt{2} > \sqrt[3]{4}$
- 45.** If $x = \sqrt{3} + \sqrt{2}$, then the value of $x^3 + x + \frac{1}{x} + \frac{1}{x^3}$, is ☑ 2015 II
(a) $10\sqrt{3}$ (b) $20\sqrt{3}$ (c) $10\sqrt{2}$ (d) $20\sqrt{2}$
- 46.** If $x = 2^{1/3} + 2^{-1/3}$, then the value of $2x^3 - 6x - 5$ is equal to ☑ 2016 I
(a) 0 (b) 1 (c) 2 (d) 3
- 47.** If $4^x 2^y = 128$ and $3^{3x} 3^{2y} - 9^{xy} = 0$, then the value of $x + y$ can be equal to ☑ 2016 I
(a) 7 (b) 5 (c) 3 (d) 1
- 48.** If $x = \frac{\sqrt{a+2b} + \sqrt{a-2b}}{\sqrt{a+2b} - \sqrt{a-2b}}$, then $bx^2 - ax + b$ is equal to (given that $b \neq 0$) ☑ 2016 I
(a) 0 (b) 1 (c) ab (d) $2ab$

> PREVIOUS YEARS' QUESTIONS

- 33.** If $a^x = b$, $b^y = c$ and $xyz = 1$, then what is the value of c^z ? ☑ 2012 I
(a) a (b) b (c) ab (d) a/b
- 34.** If $196x^4 = x^6$, then x^3 is equal to which one of the following? ☑ 2012 I
(a) $\frac{x^6}{14}$ (b) $14x^4$ (c) $\frac{x^2}{14}$ (d) $14x^2$
- 35.** If $\sqrt{10 + \sqrt[3]{x}} = 4$, then what is the value of x ? ☑ 2012 I
(a) 150 (b) 216 (c) 316 (d) 450
- 36.** The expression $[(\sqrt{2})^{\sqrt{2}}]^{\sqrt{2}}$ gives ☑ 2013 I
(a) a natural number
(b) an integer and not a natural number
(c) a rational number but not an integer
(d) a real number but not a rational number

ANSWERS

1		2		3		4		5		6		7		8		9		10	
11		12		13		14		15		16		17		18		19		20	
21		22		23		24		25		26		27		28		29		30	
31		32		33		34		35		36		37		38		39		40	
41		42		43		44		45		46		47		48					

HINTS AND SOLUTIONS

1. (c) Given, $\sqrt{\frac{x}{49}} = \frac{4}{7}$ or $\frac{\sqrt{x}}{7} = \frac{4}{7}$

Now, on squaring both sides, we get

$$(\sqrt{x})^2 = (4)^2 \Rightarrow x = 16$$

Hence, the value of x is 16.

2. (c) The digits 2, 3, 7 and 8 cannot be at the unit place of a perfect square number.

3. (b) $\sqrt{0.0001} = 0.01$ which is a rational number.

4. (b) If m and n are natural numbers, then $\frac{m}{n}$ is irrational unless n is m th power of an integer.

5. (a) Let fraction be x , then

$$x^2 = 227.798649$$

$$\Rightarrow x = \sqrt{227.798649} = 15.093$$

6. (a) Given, $x = \sqrt{3018 + \sqrt{36 + \sqrt{169}}}$

$$= \sqrt{3018 + \sqrt{36 + 13}}$$

$$= \sqrt{3018 + \sqrt{49}} = \sqrt{3018 + 7}$$

$$= \sqrt{3025} = 55$$

Hence, the value of x is 55.

7. (c) $\sqrt[3]{\sqrt{x}} = x/5$

On cubing both sides, we get

$$\sqrt{x} = x^3/5^3$$

On squaring both sides, we get

$$x = x^6/5^6 \Rightarrow 5^6 = \frac{x^6}{x} = x^5$$

$$\Rightarrow x^5 = 5^6 \Rightarrow x = \sqrt[5]{5^6}$$

8. (d) $\sqrt{29.16} + \sqrt{0.2916} + \sqrt{0.002916}$
 $\quad\quad\quad + \sqrt{0.00002916}$

$$= 5.4 + 0.54 + 0.054 + 0.0054$$

$$= 5.9994$$

9. (b) $(\sqrt{5})^{-5/2} \times (\sqrt{5})^{-3/2}$

$$= (\sqrt{5})^{-5/2 - 3/2} = (\sqrt{5})^{-8/2}$$

$$= (\sqrt{5})^{-4} = \left(\frac{1}{\sqrt{5}}\right)^4$$

$$= \left(\frac{1}{5}\right)^{(1/2) \times 4} = \left(\frac{1}{5}\right)^2 = \frac{1}{25}$$

10. (b) Given, $a = 3, b = 9$ and $c = 10$

$$\therefore \sqrt{13+a} + \sqrt{112+b} + \sqrt{c-1}$$

$$= \sqrt{13+3} + \sqrt{112+9} + \sqrt{10-1}$$

$$= \sqrt{16} + \sqrt{121} + \sqrt{9} = 4 + 11 + 3 = 18$$

11. (a) Here, $\sqrt{\frac{x}{z}} = \frac{0.9 \times 0.09}{\sqrt{0.9 \times 0.09}}$

$$= \frac{\sqrt{0.9 \times 0.09} \times \sqrt{0.9 \times 0.09}}{\sqrt{0.9 \times 0.09}}$$

$$= \sqrt{0.9 \times 0.09}$$

$$\Rightarrow \left(\sqrt{\frac{x}{z}}\right)^2 = (\sqrt{0.9 \times 0.09})^2$$

$$\Rightarrow \frac{x}{z} = 0.081$$

12. (d) Here, $\sqrt{2\sqrt{2\sqrt{2\sqrt{2\sqrt{2}}}}}$

$$= \sqrt{2\sqrt{2\sqrt{2\sqrt{2 \times 2^{1/2}}}}} = \sqrt{2\sqrt{2\sqrt{2 \times 2^{3/4}}}}$$

$$= \sqrt{2\sqrt{2 \times 2^{7/8}}} = \sqrt{2 \times 2^{15/16}} = 2^{31/32}$$

13. (c) Here, $\frac{\sqrt{7} + \sqrt{5}}{\sqrt{7} - \sqrt{5}}$

$$= \frac{\sqrt{7} + \sqrt{5}}{\sqrt{7} - \sqrt{5}} \times \frac{\sqrt{7} + \sqrt{5}}{\sqrt{7} + \sqrt{5}} = \frac{(\sqrt{7} + \sqrt{5})^2}{7 - 5}$$

$$= \frac{7 + 5 + 2\sqrt{35}}{2} = \frac{12 + 2\sqrt{35}}{2}$$

$$[\because (a+b)^2 = a^2 + b^2 + 2ab]$$

$$= 6 + \sqrt{35} = 6 + 5.9160 = 11.9160$$

14. (b) $\sqrt{343 + \sqrt{307 + \sqrt{273 + \sqrt{241 + \sqrt{225}}}}}$

$$= \sqrt{343 + \sqrt{307 + \sqrt{273 + \sqrt{241 + 15}}}}$$

$$= \sqrt{343 + \sqrt{307 + \sqrt{273 + 16}}}$$

$$= \sqrt{343 + \sqrt{307 + 17}}$$

$$= \sqrt{343 + 18} = \sqrt{361} = 19$$

15. (c) $\frac{1}{\sqrt{9-\sqrt{8}}} - \frac{1}{\sqrt{8-\sqrt{7}}} + \frac{1}{\sqrt{7-\sqrt{6}}}$
 $\quad\quad\quad - \frac{1}{\sqrt{6-\sqrt{5}}} + \frac{1}{\sqrt{5-\sqrt{4}}}$

$$= (\sqrt{9} + \sqrt{8}) - (\sqrt{8} + \sqrt{7})$$

$$+ (\sqrt{7} + \sqrt{6}) - (\sqrt{6} + \sqrt{5})$$

$$+ (\sqrt{5} + \sqrt{4})$$

[on rationalisation]

$$= \sqrt{9} + \sqrt{4} = 3 + 2 = 5$$

16. (d) $\sqrt{9-2\sqrt{14}} = \sqrt{7+2-2 \times \sqrt{7} \times \sqrt{2}}$
 $= \sqrt{(\sqrt{7}-\sqrt{2})^2} = \sqrt{7}-\sqrt{2}$

17. (a) Given, $p^x = r^y = m$ and

$$r^w = p^z = n$$

Now, $p^x = r^y$

On multiply power with w on both sides, we get

$$(p^x)^w = (r^y)^w \Rightarrow p^{xw} = r^{yw}$$

$$\Rightarrow p^{xw} = (r^w)^y \quad \dots(i)$$

On putting $r^w = p^z$ in Eq. (i), we get

$$p^{xw} = (p^z)^y \Rightarrow p^{xw} = p^{zy}$$

On comparing both sides, we get

$$\therefore xw = zy$$

18. (c) Given, $a^x = b^y = c^z = k$ [Let]

$$\Rightarrow a = k^{1/x}, b = k^{1/y}$$

and

$$c = k^{1/z}$$

$$\therefore abc = k^{\frac{1}{x} + \frac{1}{y} + \frac{1}{z}}$$

$$\Rightarrow 1 = k^{\frac{1}{x} + \frac{1}{y} + \frac{1}{z}} = k^0$$

[$\because abc = 1$, given]

On comparing both sides, we get

$$\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 0$$

$$\Rightarrow xy + yz + zx = 0$$

$$\begin{aligned}
 19. (b) \quad a &= \frac{\sqrt{5}+1}{\sqrt{5}-1} \times \frac{\sqrt{5}+1}{\sqrt{5}+1} \\
 &= \frac{(\sqrt{5}+1)^2}{5-1} = \frac{3+\sqrt{5}}{2} \\
 b &= \frac{\sqrt{5}-1}{\sqrt{5}+1} \times \frac{\sqrt{5}-1}{\sqrt{5}-1} \\
 &= \frac{(\sqrt{5}-1)^2}{5-1} = \frac{3-\sqrt{5}}{2} \\
 \therefore a^2 + b^2 &= \left(\frac{3+\sqrt{5}}{2}\right)^2 + \left(\frac{3-\sqrt{5}}{2}\right)^2 \\
 &= \frac{2(9+5)}{4} = 7
 \end{aligned}$$

$$\begin{aligned}
 \text{Also, } ab &= \frac{\sqrt{5}+1}{\sqrt{5}-1} \times \frac{\sqrt{5}-1}{\sqrt{5}+1} = 1 \\
 \therefore \frac{a^2 + ab + b^2}{a^2 - ab + b^2} &= \frac{(a^2 + b^2) + ab}{(a^2 + b^2) - ab} \\
 &= \frac{7+1}{7-1} = \frac{4}{3}
 \end{aligned}$$

$$\begin{aligned}
 20. (d) \quad 3\sqrt{5} + \sqrt{125} &= 17.88 \\
 \Rightarrow 3\sqrt{5} + 5\sqrt{5} &= 17.88 \\
 \Rightarrow 8\sqrt{5} &= 17.88 \\
 \Rightarrow \sqrt{5} &= \frac{17.88}{8} = 2.235 \\
 \therefore \sqrt{80} + 6\sqrt{5} &= 4\sqrt{5} + 6\sqrt{5} \\
 &= 10\sqrt{5} = 10 \times 2.235 = 22.35
 \end{aligned}$$

$$\begin{aligned}
 21. (a) \quad N &= 2^{0.15} \\
 \Rightarrow N &= (2)^{3/20} \Rightarrow (N)^b = (2)^{3b/20} \\
 \text{But, } N^b &= 16 \\
 \therefore 16 &= (2)^{3b/20} \Rightarrow 2^4 = (2)^{3b/20} \\
 \Rightarrow 4 &= \frac{3b}{20} \Rightarrow b = \frac{80}{3}
 \end{aligned}$$

$$\begin{aligned}
 22. (d) \quad \text{Since } 15 &= 3 \times 5 = 3 \times 6^B \\
 \text{As, } 3 &= \frac{6}{2} = \frac{6^1}{6^A} = 6^{1-A} \\
 \therefore 15 &= 3 \times 6^B = 6^{1-A} \times 6^B \\
 \Rightarrow 6^Q &= 6^{1-A} \times 6^B \Rightarrow 6^Q = 6^{1-A+B} \\
 \Rightarrow Q &= 1-A+B
 \end{aligned}$$

$$\begin{aligned}
 23. (a) \quad \text{LCM of } 4, 6, 12 &= 12 \\
 \sqrt[4]{3} &= \sqrt[12]{3^3} = \sqrt[12]{27} \\
 \sqrt[6]{10} &= \sqrt[12]{10^2} = \sqrt[12]{100} \\
 \sqrt[12]{25} &= \sqrt[12]{25}
 \end{aligned}$$

$$\begin{aligned}
 \text{Clearly, } \sqrt[12]{100} &> \sqrt[12]{27} > \sqrt[12]{25} \\
 \text{Thus, } \sqrt[6]{10} &> \sqrt[4]{3} > \sqrt[12]{25}
 \end{aligned}$$

$$\begin{aligned}
 24. (c) \quad \text{Given, } (3.7)^x &= (0.037)^y = 10000 \\
 \Rightarrow (3.7)^x &= 10^4 \quad \text{and} \quad (0.037)^y = 10^4 \\
 \Rightarrow 37 &= 10^{4/x+1} \quad \text{and} \quad 37 = 10^{4/y+3} \\
 \Rightarrow 10^{4/x+1} &= 10^{4/y+3}
 \end{aligned}$$

$$\begin{aligned}
 \text{On comparing both sides, we get} \\
 \frac{4}{x} + 1 &= \frac{4}{y} + 3
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow \frac{4}{x} - \frac{4}{y} &= 3 - 1 \Rightarrow \frac{1}{x} - \frac{1}{y} = \frac{1}{2} \\
 \text{Hence, the value of } \frac{1}{x} - \frac{1}{y} &\text{ is } \frac{1}{2}.
 \end{aligned}$$

$$\begin{aligned}
 25. (d) \quad \text{Here, the greatest six-digit number} \\
 &= 999999
 \end{aligned}$$

$$\begin{array}{r}
 999 \\
 9 \overline{) 99 \, 99 \, 99} \\
 \underline{81} \\
 189 \\
 \times 9 \\
 \hline
 1989 \\
 \times 9 \\
 \hline
 19899 \\
 \times 9 \\
 \hline
 17901 \\
 \hline
 1998
 \end{array}$$

$$\begin{aligned}
 \text{Here, the greatest number of six digit} \\
 \text{which is perfect square} \\
 &= 999999 - 1998 = 998001
 \end{aligned}$$

$$\begin{aligned}
 26. (c) \quad \text{We have,} \\
 26244 &= 2 \times 2 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3 \\
 \text{So, to get a perfect cube,} \\
 26244 &\text{ must be divided by } 2 \times 2 \times 3 \times 3 \\
 \text{i.e., } 36.
 \end{aligned}$$

$$\begin{aligned}
 27. (d) \quad \text{We know that, } (42)^2 &= 1764 \quad \text{and} \\
 (43)^2 &= 1849 \\
 \text{Since, } 1764 < 1780 < 1849 \\
 \text{Hence, the smallest number that must be} \\
 \text{added to 1780 is } (1849 - 1780), \text{ i.e. } 69.
 \end{aligned}$$

$$\begin{aligned}
 28. (c) \quad \text{Given, total number of trees} &= 17956 \\
 \therefore \text{Number of trees in each row} \\
 &= \sqrt{17956} = 134
 \end{aligned}$$

$$\begin{aligned}
 29. (c) \quad \text{Total rupees collected} &= ₹ 32.49 \\
 &= 32.49 \times 100 \text{ paise} = 3249 \text{ paise} \\
 \therefore \text{Number of member in the group} \\
 &= \sqrt{3249} = 57
 \end{aligned}$$

$$\begin{aligned}
 30. (a) \quad \text{Here, number of soldier arranged} \\
 &= 8160 - 60 = 8100 \\
 \text{As number of soldier in each row is} \\
 \text{equal to number of row.} \\
 \text{So, number of soldier in each row} \\
 &= \sqrt{8100} = 90
 \end{aligned}$$

$$\begin{aligned}
 31. (c) \quad \text{After Ist hit, height of the ball} \\
 &= \frac{1}{2} (64) \\
 \text{After IInd hit, height of the ball} \\
 &= \left(\frac{1}{2}\right)^2 (64)
 \end{aligned}$$

$$\begin{aligned}
 \text{After 16th hit, height of the ball} \\
 &= \left(\frac{1}{2}\right)^{16} (64) \\
 &= \frac{1}{2^{16}} (2^6) = 2^{-10} \text{ m}
 \end{aligned}$$

$$\begin{aligned}
 32. (b) \quad \text{Given, } x &= \frac{\sqrt{a+3b} + \sqrt{a-3b}}{\sqrt{a+3b} - \sqrt{a-3b}} \\
 \text{By rationalising, we get} \\
 x &= \frac{(\sqrt{a+3b})^2 + (\sqrt{a-3b})^2}{(\sqrt{a+3b})^2 - (\sqrt{a-3b})^2} \\
 &\quad + \frac{2\sqrt{(a+3b)(a-3b)}}{(\sqrt{a+3b})^2 - (\sqrt{a-3b})^2} \\
 x &= \frac{a+3b + a-3b + 2\sqrt{a^2 - 9b^2}}{a+3b - a+3b} \\
 x &= \frac{2a + 2\sqrt{a^2 - 9b^2}}{a+3b - a+3b} \\
 &= \frac{(a + \sqrt{a^2 - 9b^2})}{(3b)}
 \end{aligned}$$

$$3bx^2 - 2ax + 3b = 0$$

$$\begin{aligned}
 33. (a) \quad \text{Given, } a^x &= b, \quad b^y = c \quad \text{and} \quad xyz = 1 \\
 \text{Now, } a^x &= b \\
 \text{On multiplying both sides by } y \text{ in} \\
 \text{power, we get} \\
 (a^x)^y &= (b)^y \Rightarrow a^{xy} = b^y \\
 \text{Here, } b^y &= c \Rightarrow a^{xy} = c \\
 \text{Again, multiply with } z \text{ in power, we get} \\
 (a^{xy})^z &= c^z \Rightarrow a^{xyz} = c^z \\
 \text{Given, } xyz &= 1 \Rightarrow a = c^z \\
 \therefore c^z &= a \\
 \text{Hence, the value of } c^z &\text{ is } a.
 \end{aligned}$$

$$34. (d) \quad 196x^4 = x^6, \quad 196 = \frac{x^6}{x^4} = x^2$$

$$\begin{aligned}
 \Rightarrow x &= \sqrt{196} = 14 \\
 \Rightarrow x^3 &= 14x^2
 \end{aligned}$$

$$\begin{aligned}
 35. (b) \quad \text{Given, } \sqrt{10 + \sqrt[3]{x}} &= 4 \\
 \text{On squaring both sides, we get} \\
 10 + \sqrt[3]{x} &= 16 \\
 \Rightarrow \sqrt[3]{x} &= 16 - 10 = 6 \\
 \text{Now, on cubing both sides, we get} \\
 x &= (6)^3 = 216 \\
 \text{Hence, the value of } x &\text{ is } 216.
 \end{aligned}$$

$$\begin{aligned}
 36. (d) \quad \text{Expression} &= [(\sqrt{2})^{\sqrt{2}}]^{\sqrt{2}} \\
 &= (\sqrt{2})^{(\sqrt{2})^{\sqrt{2}}} = (\sqrt{2})^{(2)^{\frac{1}{\sqrt{2}}}} \\
 &= (2)^{\frac{1}{2} \times 2^{\frac{1}{\sqrt{2}}}} = (2)^{(2)^{\left(\frac{1}{\sqrt{2}} - 1\right)}}
 \end{aligned}$$

which denotes a real number but not a rational number.

$$\begin{aligned}
 37. (d) \quad \text{Given, } 16 \times 8^{n+2} &= 2^m \\
 \Rightarrow (2^4) \times 2^{3(n+2)} &= 2^m \\
 \Rightarrow 2^{(4+3n+6)} &= 2^m \Rightarrow 2^{(3n+10)} = 2^m \\
 \text{On comparing, we get} \\
 3n + 10 &= m \\
 \Rightarrow m &= 3n + 10
 \end{aligned}$$